

Model 4 - Differentially Private Recommender

DP-SASRec using user-level DP-SGD

Goal: quantify exactly how recommendation quality changes as formal privacy protection becomes stronger.

What the model is

Model 4 is the differentially private version of the project's sequential recommender. It keeps centralized training, but changes how learning is performed so that no single user can have an unrestricted influence on the trained model.

The proposed implementation is DP-SASRec using user-level DP-SGD. Each user contribution is clipped to a maximum gradient norm, calibrated Gaussian noise is added before the update, and a privacy accountant reports the final (epsilon, delta) differential-privacy guarantee.

A smaller epsilon means stronger privacy. Stronger privacy usually requires more noise, which can reduce recommendation quality. That trade-off is exactly what Model 4 is designed to measure.

How it differs from the other models

Model	Main idea	Primary privacy mechanism
Model 1 - Centralized	Train normally on pooled user histories	No special privacy mechanism
Model 2 - Federated	Keep raw histories local and aggregate updates	Data locality + secure aggregation
Model 3 - On-device	Personalize and serve recommendations locally	Data remains on device
Model 4 - DP	Train centrally while limiting and randomizing user contributions	Formal (epsilon, delta)-DP guarantee

The key distinction is that federated learning and secure aggregation are not the same as differential privacy. Model 2 mainly changes where data and updates are visible. Model 4 changes the training algorithm itself and provides a numeric privacy budget.

Why the team needs it

Model 4 is the clearest model for answering the project's central research question: how much recommendation quality is lost when privacy protection becomes stronger?

Because epsilon is a controllable privacy parameter, we can train the same recommender at several privacy budgets and measure NDCG@10 and Recall@10 at each point. The final result is a privacy-utility frontier rather than a single accuracy number.

The approach in one paragraph

Start from the same SASRec-style sequential backbone used as the project's common reference. Sort each user's interactions chronologically and represent the user as one sequence. During training, compute each user's gradient separately, clip it to a fixed norm, aggregate the clipped gradients, add Gaussian noise, and update the

model. A privacy accountant tracks cumulative privacy loss over training and reports the final (epsilon, delta) guarantee.

User sequence	SASRec	Per-user gradient	Clip	Add Gaussian noise	Private update	DP-SASRec
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One decision to make before any code is written

The most important design decision is the privacy unit: are we protecting one interaction, or one entire user?

Privacy unit	Meaning	Implication
Interaction-level DP	One click/rating is protected	A user may contribute many protected records; weaker user-level interpretation
User-level DP	A complete user history is one protected unit	More meaningful for recommendation, but requires careful batching and accounting

Recommended choice: user-level DP. One training sample should correspond to one chronologically ordered user sequence, so per-sample clipping in DP-SGD corresponds to per-user clipping.

How to implement it

1. Start from the shared SASRec implementation

The core comparison should be non-private SASRec versus DP-SASRec. Keeping the architecture, dataset, preprocessing, and evaluation fixed makes the privacy cost easier to isolate.

2. Use one user sequence as one DP sample

For MovieLens-1M, group interactions by user, sort by timestamp, and use the same chronological split as the other models. A typical split is earlier interactions for training, the second-to-last interaction for validation, and the last interaction for testing.

Example: [A, B, C] -> training history; D -> validation; E -> test.

3. Apply per-user gradient clipping

For user i with gradient $\mathbf{g}_i$, clip the norm to at most C :

$$\mathbf{g}_{i_bar} = \mathbf{g}_i * \min(1, C / ||\mathbf{g}_i||_2)$$

This bounds the maximum effect of any single user on the model update.

4. Add Gaussian noise

$$\mathbf{g_tilde} = (1/B) [\sum_i \mathbf{g}_{i_bar} + N(0, \sigma^2 C^2 I)]$$

The noise obscures individual contributions. The combination of clipping, noise, and privacy accounting provides the formal privacy guarantee.

5. Use Opacus for DP-SGD

The practical implementation path is PyTorch + Opacus. Opacus provides per-sample gradients, clipping, Gaussian noise, and privacy accounting through its PrivacyEngine.

- target_epsilon and target_delta
- max_grad_norm
- noise_multiplier
- number of epochs and sampling rate

6. Run a privacy-budget sweep

Do not report only one DP model. A useful initial sweep is epsilon in {2, 5, 8, 10}, plus a no-DP reference. The exact values can be finalized after compatibility and compute tests.

Main research paper

Ding et al. (2024), "Delving into Differentially Private Transformer," ICML 2024, is a strong reference for the implementation direction because it studies Transformer training under differential privacy and includes sequential recommendation experiments.

The paper identifies two Transformer-specific problems under DP training:

- Attention distraction caused by DP noise.
- High memory/compute cost of efficient per-sample gradient clipping.

It proposes Re-Attention to recover utility under DP noise and Phantom Clipping to make DP training more efficient. The paper evaluates MovieLens with epsilon values 5, 8, and 10, making it directly relevant to our planned privacy sweep.

Recommended scope: implement standard DP-SASRec with DP-SGD first. If time permits, add Re-Attention as an enhancement or ablation.

Supporting paper: DIPSGNN

Hu and Fang's DIPSGNN is useful related work because it directly studies differential privacy in sequential recommendation. However, it uses a noisy graph neural network rather than SASRec. Therefore it is better treated as a conceptual/sequential-DP reference than as the core implementation if our goal is to isolate the effect of privacy while keeping the backbone comparable.

What we measure and report

Metric / setting	Why it is needed
NDCG@10	Primary top-K ranking quality metric
Recall@10 / Hit@10	Whether the correct item appears in the top recommendations
epsilon	Strength of the privacy guarantee
delta	Small failure-probability parameter
Noise multiplier	Amount of Gaussian noise used by DP-SGD
Clipping norm	Maximum contribution allowed per user
Training time	DP-SGD can be substantially slower than normal training
GPU memory	Per-user gradients increase memory cost
Cold-start performance	Privacy may hurt sparse users disproportionately
Membership-inference success / AUC	Measures whether private training reduces leakage

Data source

Primary dataset: MovieLens-1M

MovieLens-1M should remain the common primary benchmark so Model 4 is evaluated on the same user histories and fixed splits as the other models. In the team materials, the SASRec-preprocessed MovieLens setup is described as approximately 6,040 users, 3,416 movies, and 1.0M interactions.

For Model 4, the important design point is that a user history can naturally be represented as one DP sample. All models should share the same filtering rules, vocabulary, maximum history length where possible, chronological split, and evaluation code.

Required resources and dependencies

Component	Recommended choice
Model code	PyTorch SASRec implementation
Core libraries	PyTorch, NumPy, pandas
Differential privacy	Opacus
Privacy accounting	Opacus RDP/PRV accountant
Primary dataset	MovieLens-1M
Training compute	Department GPU Lab
Experiment logging	Same tracking setup as the other models
Optional advanced method	Re-Attention / Phantom Clipping

No special privacy hardware is required. The main extra system cost relative to non-private SASRec is memory and training time because DP-SGD needs per-user gradients before aggregation.

How we compare against the reference model

The most important controlled comparison is non-private SASRec versus DP-SASRec. Keep the dataset, preprocessing, chronological split, maximum sequence length, embedding size where possible, candidate-ranking procedure, evaluation code, and random seeds identical.

Then report both the absolute and relative privacy cost:

$$\text{Absolute loss} = \text{NDCG_non-private} - \text{NDCG_DP}$$

$$\text{Relative loss (\%)} = (\text{NDCG_non-private} - \text{NDCG_DP}) / \text{NDCG_non-private} * 100$$

What success looks like

Success does not mean matching the non-private recommender. Some utility loss is expected if privacy is doing real work. A successful result is being able to state, at a specific privacy budget, exactly how much ranking quality changed and whether leakage was reduced.

Example target statement: "At epsilon = 5, user-level DP reduced NDCG@10 by X% relative to non-private SASRec while reducing membership-inference success to Y%."

Open items for the team

- Confirm whether the common-backbone experiment will use SASRec for Models 1-4.
- Finalize the epsilon sweep and delta value.
- Assign the membership-inference attack used across all models.
- Confirm whether Amazon Beauty is a secondary dataset or optional stretch goal.
- Decide whether Re-Attention is core scope or an optional enhancement.
- Test SASRec compatibility with Opacus before locking the implementation.

References used for this brief

Ding, J. et al. (2024). Delving into Differentially Private Transformer. Proceedings of ICML 2024.

Hu, L. & Fang, H. Towards Differential Privacy in Sequential Recommendation: A Noisy Graph Neural Network Approach (DIPSGNN).

Kang, W.-C. & McAuley, J. (2018). Self-Attentive Sequential Recommendation. ICDM 2018.

Opacus documentation. PrivacyEngine and DP-SGD training utilities.

Team project plan: Topic 22 - Personalization and Recommendation Under Privacy Constraints.

Team Model 3 research brief: On-Device Recommender.